

Assignment 1; Solutions

- [1] Let $E \subseteq \mathbb{R}$ be nonempty. We say that $t \in \mathbb{R}$ is a *lower bound* of E if $t \leq x$ for all $x \in E$. We say that E is *bounded below* if there exists a lower bound of E . We say that $t \in \mathbb{R}$ is a *greatest lower bound*, or *infimum*, of E if it is a lower bound of E and if $t \geq s$ for any lower bound s of E . If E has an infimum, it is denoted $\inf E$.
- [a] Let E be a nonempty subset of $\mathbb{R}$ which is bounded below. Find a way to use the completeness axiom to prove that $\inf E$ exists. *Hint:* Consider the subset $-E = \{-x : x \in E\}$.
- [b] Let $\emptyset \neq A \subseteq B \subseteq \mathbb{R}$. Suppose B is bounded below (so $\inf B$ exists). Prove that A is bounded below (so $\inf A$ exists) and that $\inf A \geq \inf B$.
- [c] Let A, B be nonempty subsets of $\mathbb{R}$, both bounded below (so both $\inf A$ and $\inf B$ exist.) Define

$$A + B = \{a + b : a \in A, b \in B\}.$$

Prove that $A + B$ is bounded below and that $\inf(A + B) = \inf A + \inf B$.

Solution.

- [a] Since E is bounded from below, there exists $t \in \mathbb{R}$ such that $x \geq t$ for all $x \in E$. But then $-x \leq -t$ for all $x \in E$. Define $-E = \{-x : x \in E\}$. This is a nonempty subset of $\mathbb{R}$ which is bounded above by $-t$. Thus $\sup(-E)$ exists. That is, $-x \leq \sup(-E)$ for all $x \in E$, so $x \geq -\sup(-E)$ for all $x \in E$. Hence $-\sup(-E)$ is a lower bound of E .
- Let $\delta > 0$, so $-\sup(-E) + \delta > -\sup(-E)$. Thus $\sup(-E) - \delta < \sup(-E)$, so there exists $x \in E$ such that $\sup(-E) - \delta < -x \leq \sup(-E)$. This means $-\sup(-E) + \delta > x \geq -\sup(-E)$, so $-\sup(-E) + \delta$ is not a lower bound for E . Hence $-\sup(-E)$ is the greatest lower bound. That is, $\inf E$ exists, and $\inf E = -\sup(-E)$.
- [b] Let t be a lower bound for B , so $b \geq t$ for all $b \in B$. Since $A \subseteq B$, we get $a \geq t$ for all $a \in A$. Thus t is a lower bound for A , so A is bounded below. Since $\inf B$ is a lower bound for B , and thus for A , it must be less than or equal to the greatest lower bound of A . That is, $\inf B \leq \inf A$.
- [c] We have $\inf A \leq a$ for all $a \in A$, and $\inf B \leq b$ for all $b \in B$. Thus $\inf A + \inf B \leq a + b$ for all $a + b \in A + B$. Hence $A + B$ is bounded below by $\inf A + \inf B$, so $\inf(A + B)$ exists and $\inf A + \inf B \leq \inf(A + B)$.
- Let $\delta > 0$. There exists $a \in A$ such that $a < \inf A + \frac{1}{2}\delta$, and there exists $b \in B$ such that $b < \inf B + \frac{1}{2}\delta$. Thus $a + b < \inf A + \inf B + \delta$, so $\inf A + \inf B + \delta$ is not a lower bound for $A + B$. That is, any real number strictly larger than $\inf A + \inf B$ is not a lower bound for $A + B$. Since $\inf(A + B)$ is a lower bound for $A + B$, we must have $\inf(A + B) \leq \inf A + \inf B$.
- Putting these two inequalities together gives $\inf(A + B) = \inf A + \inf B$. Alternatively, we could use part [a] and the analogous result for $\sup$, via

$$\begin{aligned} \inf(A + B) &= -\sup(-(A + B)) = -\sup((-A) + (-B)) = -(\sup(-A) + \sup(-B)) \\ &= -\sup(-A) - \sup(-B) = \inf A + \inf B. \end{aligned}$$

- [2] Let x, y be two real numbers such that $y - x > 1$. (In particular, $x < y$.)
- [a] If x is positive, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.
- [b] If x is negative and y is positive, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.
- [c] If y is negative, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.

Solution.

- [a] Since $x > 0$, by the Archimedean property, there exists an integer k such that $k - 1 \leq x < k$. Suppose that $y \leq k$. Then we have $k - 1 \leq x < y \leq k$, so $y - x \leq 1$, which contradicts $y - x > 1$. Thus we must have $y > k$, so $x < k < y$.

- [b] If x is negative and y is positive, we can take $k = 0 \in \mathbb{Z}$, and then $x < k < y$.
- [c] Suppose that y is negative, so x is also negative. Then $-x, -y$ are both positive, and $(-x) - (-y) = y - x > 1$. (In particular, $-y < -x$.) Applying part [a], we find an integer $m \in \mathbb{Z}$ such that $-y < m < -x$. Multiplying through by -1 , we get $x < -m < y$, so we can take $k = -m \in \mathbb{Z}$.

[3] The parts of this problem are *not* related.

- [a] Let $x \in \mathbb{R}$. Consider the set $E = \{q \in \mathbb{Q} : q < x\} \subset \mathbb{Q} \subset \mathbb{R}$. Show that E is nonempty and bounded above, so $\sup E$ exists. Show also that $\sup E = x$.
- [b] Let $x, y \in \mathbb{R}$ with $x < y$. Prove that there exists an *irrational* (not rational) number z such that $x < z < y$. *Hint:* You may use the fact that $\sqrt{2}$ is irrational.

Solution.

- [a] Since $x - 1 < x$, there exists a rational number p with $x - 1 < p < x$, so E is nonempty. Clearly, $q < x$ for all $q \in E$, so x is an upper bound of E . Thus E is bounded above, so $\sup E$ exists, and $\sup E \leq x$.
- Let $\delta > 0$. Then $x - \delta < x$, so there exists $p \in \mathbb{Q}$ with $x - \delta < p < x$. In particular $p \in E$. Hence $x - \delta$ cannot be an upper bound for E . (That is, any number strictly smaller than x is not an upper bound for E .) Since $\sup E$ is an upper bound for E , we get $\sup E \geq x$. Thus $\sup E = x$.
- [b] Since $\sqrt{2} > 0$, we have $\frac{1}{\sqrt{2}}x < \frac{1}{\sqrt{2}}y$. There exists a rational number q such that $\frac{1}{\sqrt{2}}x < q < \frac{1}{\sqrt{2}}y$, so $x < \sqrt{2}q < y$. But $\sqrt{2}q$ must be irrational, because if it was a rational number p , then $\sqrt{2}q = p$ implies that $\sqrt{2} = \frac{p}{q}$ would be rational, but it is not.

[4] Let $C > 0$ be a fixed positive number.

- [a] Let $x \geq 0$, and suppose that for any $\varepsilon > 0$, we have $x < C\varepsilon$. Show that for any $\eta > 0$, we have $x < \eta$.
- [b] In the situation of part [a], show that we must have $x = 0$. *Hint:* Find a way to use the Archimedean property.

Solution.

- [a] Let $\eta > 0$. Define $\varepsilon = \frac{\eta}{C}$, which is also positive. Then by assumption, we have $x < C\varepsilon$, but $C\varepsilon = \eta$, so $x < \eta$.
- [b] We prove this by contradiction. Suppose $x \neq 0$. Then since $x \geq 0$, we must have $x > 0$, so $\frac{1}{x} > 0$. By the Archimedean property, there exists $n \in \mathbb{N}$ such that $n > \frac{1}{x}$, so $x > \frac{1}{n}$. But this contradicts the assumption that x is strictly less than any positive number.

[5] In this problem you will give an *alternative proof* of the Cauchy–Schwarz inequality. The inequality says that

$$-\|\mathbf{x}\|\|\mathbf{y}\| \leq \langle \mathbf{x}, \mathbf{y} \rangle \leq \|\mathbf{x}\|\|\mathbf{y}\|.$$

Both inequalities are trivially seen to be equalities if either $\mathbf{x} = \mathbf{0}$ or $\mathbf{y} = \mathbf{0}$, so we suppose that both vectors are nonzero.

By consider the norm of the vector $\mathbf{z} = \frac{\mathbf{x}}{\|\mathbf{x}\|} \pm \frac{\mathbf{y}}{\|\mathbf{y}\|}$, show that

- [a] both inequalities are true,
- [b] the first inequality becomes an equality if and only if $\mathbf{y} = t\mathbf{x}$ for some $t < 0$,
- [c] the second inequality becomes an equality if and only if $\mathbf{y} = t\mathbf{x}$ for some $t > 0$.

Solution. Note that

$$\begin{aligned}\|\mathbf{z}\|^2 &= \mathbf{z} \cdot \mathbf{z} = \left(\frac{\mathbf{x}}{\|\mathbf{x}\|} \pm \frac{\mathbf{y}}{\|\mathbf{y}\|} \right) \cdot \left(\frac{\mathbf{x}}{\|\mathbf{x}\|} \pm \frac{\mathbf{y}}{\|\mathbf{y}\|} \right) \\ &= \frac{\mathbf{x}}{\|\mathbf{x}\|} \cdot \frac{\mathbf{x}}{\|\mathbf{x}\|} \pm 2 \frac{\mathbf{x}}{\|\mathbf{x}\|} \cdot \frac{\mathbf{y}}{\|\mathbf{y}\|} + \frac{\mathbf{y}}{\|\mathbf{y}\|} \cdot \frac{\mathbf{y}}{\|\mathbf{y}\|} \\ &= 2 \pm 2 \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}.\end{aligned}$$

Since $\|\mathbf{z}\|^2 \geq 0$ with equality if and only if $\mathbf{z} = \mathbf{0}$, we deduce that

$$\begin{aligned}0 &\leq 2 \pm 2 \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} \\ -2 &\leq \pm 2 \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} \\ \|\mathbf{x}\| \|\mathbf{y}\| &\geq \mp \mathbf{x} \cdot \mathbf{y}\end{aligned}$$

with equality if and only if $\mathbf{z} = \mathbf{0}$. But the two cases of sign then precisely give

$$\begin{aligned}-\|\mathbf{x}\| \|\mathbf{y}\| &\leq \mathbf{x} \cdot \mathbf{y} && \text{with equality iff } \frac{\mathbf{x}}{\|\mathbf{x}\|} = -\frac{\mathbf{y}}{\|\mathbf{y}\|}, \\ \mathbf{x} \cdot \mathbf{y} &\leq \|\mathbf{x}\| \|\mathbf{y}\| && \text{with equality iff } \frac{\mathbf{x}}{\|\mathbf{x}\|} = \frac{\mathbf{y}}{\|\mathbf{y}\|},\end{aligned}$$

which is what we wanted to prove.

- [6] Let $E = \overline{B_r(\mathbf{x}_0)} = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_0\| \leq r\}$ be a closed ball. Prove that E is *not open*.

Solution. Let $\mathbf{x} = \mathbf{x}_0 + r\mathbf{e}_1$, where $\mathbf{e}_1 = (1, 0, \dots, 0)$. Then $\|\mathbf{x} - \mathbf{x}_0\| = \|\mathbf{e}_1\| = r$, so $\mathbf{x} \in E$. Let $\delta > 0$ be such that $\delta < r$. Consider the point $\mathbf{z} = \mathbf{x}_0 + (r + \frac{\delta}{2})\mathbf{e}_1 \in \mathbb{R}^n$. Then $\|\mathbf{z} - \mathbf{x}_0\| = r + \frac{\delta}{2} > r$, so $\mathbf{z} \in E^c$, and $\|\mathbf{z} - \mathbf{x}\| = \frac{\delta}{2} < \delta$, so $\mathbf{z} \in B_\delta(\mathbf{x})$. This shows that $B_\delta(\mathbf{x}) \cap E^c \neq \emptyset$, because $\mathbf{z}$ lies in both sets. But this means that $B_\delta(\mathbf{x})$ is not completely contained in E for any $0 < \delta < r$. Hence E is not open.

- [7] Let $F = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ be a finite subset of $\mathbb{R}^n$. Prove that F is *closed*.

Solution. Let $\mathbf{y} \in F^c$. Then $\mathbf{y} \neq \mathbf{x}_j$ for $1 \leq j \leq N$, so $d_j = \|\mathbf{y} - \mathbf{x}_j\| > 0$ for all j . Let $\delta = \min\{d_1, \dots, d_N\} > 0$. If $\|\mathbf{z} - \mathbf{y}\| < \delta$, then

$$\|\mathbf{z} - \mathbf{x}_j\| \geq \|\mathbf{x}_j - \mathbf{y}\| - \|\mathbf{y} - \mathbf{z}\| = d_j - \|\mathbf{y} - \mathbf{z}\| \geq \delta - \|\mathbf{y} - \mathbf{z}\| > 0,$$

so $\mathbf{z} \neq \mathbf{x}_j$ for any $1 \leq j \leq N$. Thus $\mathbf{z} \in F^c$. But that means $B_\delta(\mathbf{y}) \subseteq F^c$, and hence since $\mathbf{y} \in F^c$ was arbitrary, F^c is open and thus F is closed.

- [8] Let $\mathbf{a} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$, and $c \in \mathbb{R}$. Consider the subset $E = \{\mathbf{x} \in \mathbb{R}^n : \langle \mathbf{x}, \mathbf{a} \rangle < c\}$. Prove that E is open.
Hint: Observe that $|\langle \mathbf{y}, \mathbf{a} \rangle - \langle \mathbf{z}, \mathbf{a} \rangle| = |\langle \mathbf{y} - \mathbf{z}, \mathbf{a} \rangle| \leq \|\mathbf{y} - \mathbf{z}\| \|\mathbf{a}\|$ by Cauchy–Schwarz.

Solution. Let $\mathbf{y} \in E$. Thus $\langle \mathbf{y}, \mathbf{a} \rangle < c$, so $c - \langle \mathbf{y}, \mathbf{a} \rangle > 0$. Define $\delta = \|\mathbf{a}\|^{-1}(c - \langle \mathbf{y}, \mathbf{a} \rangle) > 0$, which makes sense because $\mathbf{a} \neq \mathbf{0}$. Now if $\mathbf{z} \in B_\delta(\mathbf{y})$, then $\|\mathbf{z} - \mathbf{y}\| < \delta$. Hence, by the hint, we have

$$|\langle \mathbf{y}, \mathbf{a} \rangle - \langle \mathbf{z}, \mathbf{a} \rangle| \leq \|\mathbf{y} - \mathbf{z}\| \|\mathbf{a}\| < \delta \|\mathbf{a}\| = c - \langle \mathbf{y}, \mathbf{a} \rangle.$$

But this means that

$$\langle \mathbf{z}, \mathbf{a} \rangle = \langle \mathbf{y}, \mathbf{a} \rangle - \langle \mathbf{y}, \mathbf{a} \rangle + \langle \mathbf{z}, \mathbf{a} \rangle < c - \langle \mathbf{y}, \mathbf{a} \rangle + \langle \mathbf{y}, \mathbf{a} \rangle = c,$$

so $\mathbf{z} \in E$. We have shown that $B_\delta(\mathbf{y}) \subseteq E$. Since $\mathbf{y} \in E$ was arbitrary, we deduce that E is open.

- [9] Let $C \subset \mathbb{R}^n$ be a non-empty proper *closed* subset of $\mathbb{R}^n$. Let $\mathbf{x} \in \mathbb{R}^n \setminus C$, which exists because C is a proper subset. Define the subset $E \subset \mathbb{R}$ by

$$E = \{r > 0 : B_r(\mathbf{x}) \cap C = \emptyset\}.$$

- [a] Explain why $E \neq \emptyset$
[b] Prove that there exists an upper bound for E .
[c] Thus, by the completeness axiom, $\sup E$ exists. Show that $\sup E \leq \|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{y} \in C$.

Solution.

- [a] Since C is closed, its complement $\mathbb{R}^n \setminus C$ is open. Since $\mathbf{x} \in \mathbb{R}^n \setminus C$, there exists $\delta > 0$ such that $B_\delta(\mathbf{x}) \subseteq \mathbb{R}^n \setminus C$, so $B_\delta(\mathbf{x}) \cap C = \emptyset$, and hence $\delta \in E$. Thus $E \neq \emptyset$.
[b] Since C is nonempty, there exists $\mathbf{y} \in C$. Let $d = \|\mathbf{x} - \mathbf{y}\|$, which is positive since $\mathbf{y} \neq \mathbf{x}$. For any $\varepsilon > 0$, we have $\mathbf{y} \in B_{d+\varepsilon}(\mathbf{x})$, so $B_{d+\varepsilon}(\mathbf{x}) \cap C \neq \emptyset$, since $\mathbf{y}$ is in there. Thus the real number $d + \varepsilon$ is not in E . Hence, any real number strictly larger than d is not in E , so if $r \in E$, then $r \leq d$. Therefore d is an upper bound for E .
[c] In part [b] we proved that if $\mathbf{y} \in C$, then $d = \|\mathbf{x} - \mathbf{y}\|$ is an upper bound for E . We therefore have $\sup E \leq \|\mathbf{x} - \mathbf{y}\|$ for any $\mathbf{y} \in C$.

- [10] Let $(\mathbf{x}_k)_{k \in \mathbb{N}}$ be a sequence in $\mathbb{R}^n$, and suppose that there exists $r \in (0, 1)$ and $C > 0$ such that

$$\|\mathbf{x}_{k+1} - \mathbf{x}_k\| \leq Cr^k.$$

Prove that $(\mathbf{x}_k)$ is Cauchy.

Hint: Recall that $r^{k+m-1} + r^{k+m-2} + \dots + r^k = r^k(r^{m-1} + r^{m-2} + \dots + 1) = r^k \frac{1-r^m}{1-r}$.

You may also use without proof the fact (which is easy to show using the Archimedean Property) that: since $r \in (0, 1)$, given any $\delta > 0$, there exists $N \in \mathbb{N}$ such that $k \geq N$ implies $r^k < \delta$.

Solution. By the triangle inequality, the hypothesis, and the hint, we have

$$\begin{aligned} \|\mathbf{x}_{k+m} - \mathbf{x}_k\| &= \left\| \sum_{j=0}^{m-1} (\mathbf{x}_{k+j+1} - \mathbf{x}_{k+j}) \right\| \\ &\leq \sum_{j=0}^{m-1} \|\mathbf{x}_{k+j+1} - \mathbf{x}_{k+j}\| \leq C(r^{k+m-1} + r^{k+m-2} + \dots + r^k) \\ &\leq Cr^k \frac{1-r^m}{1-r} < \frac{C}{1-r} r^k. \end{aligned}$$

Since $r \in (0, 1)$, for any $\varepsilon > 0$ we can find N such that $k \geq N$ implies that

$$r^k < \frac{\varepsilon(1-r)}{C},$$

and thus $\|\mathbf{x}_{k+m} - \mathbf{x}_k\| < \varepsilon$ for all $k \geq N$, which is what we wanted to show.